

Route-F P54: Scalar Matching, a Common Tadpole Anchor, Factorizable Wilson Terms and Finite Seesaw Thresholds

P54PQ-v2 four-dimensional mainline

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Abstract

We calculate the upper scalar hard two-point function on the existing Pati–Salam saddle, propagate the already fixed bosonic counterterms to that saddle, and evaluate a factorizable scalar-exchange Wilson response. We also calculate an exact constant-background scalar determinant subtraction at the lower broken background and implement the universal gauge/quartic finite seesaw contribution at moving mass thresholds. The principal new warning is structural: the computed heavy mass insertion has relative spectral radius 6.76, although some projected source corrections are much smaller. Its propagator expansion is not controlled at the displayed benchmark. This is not a full-model exclusion, because several matching contributions remain absent. All results below have explicit subset boundaries; complete finite matching, the fermion-inclusive light state and a physical fit remain open.

1 Fixed theory, prescription and deliverable boundary

We retain the existing renormalizable P54PQ-v2 action with $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$, its two complex symmetric UV family matrices, and its previous scalar parameter point. We neither fit a new mass formula nor add lattice scans, geometric particle labels, or family parameters. The upper tree stationary PS saddle is

$$(w_U, \sigma_U, v_{s,U}) = (1.00081030592151, 0, 0.254277697083708). \quad (1)$$

The historical broken tree vacuum is $(1, 0.1265, 0.25)$. All masses and fields in numerical tables use its ω as reference. We use DR with $d = 4 - 2\epsilon$, $\overline{\text{MS}}$, background-field Landau gauge and the *same* previous finite fixed-VEV bosonic counterterms. DR here means dimensional regularization, not dimensional reduction. Define $\mathcal{L} = 16\pi^2$.

The common counterterm reference is

$$\mu_0 = 0.5626028899853225\,\omega, \quad M_V^2 = 0.3170351789677566\,\omega^2. \quad (2)$$

All new upper terms are evaluated at μ_0 , not the nearby historical $\mu_U = 0.5630587704385367\,\omega$. One must evolve or convert the counterterms when changing their anchor. The upper field census is 55 positive physical heavy real scalars, 248 retained real PS scalar coordinates, 24 gauge Goldstones and one physical PQ direction. Retained curvatures need not be positive at this intermediate saddle. They are not integrated as unstable massive particles.

The existing P-MATCH-KJC1 note proves the first-order transport of kernel, sources and contact terms. This note supplies further diagrams and tests; it does not turn transport identities into a diagram-completeness theorem. The finite matching method is based on equality of the full and EFT amplitudes after their common IR contributions are subtracted [1, 2].

2 Actual upper scalar hard two-point function

2.1 Action derivatives, not interpolation of a spectrum

Let X be the 328-component canonical real scalar coordinate, $V(X)$ the fixed quartic potential and $H(X) = V''(X)$. At $X = x_U$ define

$$T_a = \partial_a H(x_U), \quad U_{ab} = \partial_a \partial_b H(x_U). \quad (3)$$

Because H is quadratic in X , for every real direction q and every nonzero step s the polynomial identities are

$$T_q = \frac{H(x_U + sq) - H(x_U - sq)}{2s}, \quad (4)$$

$$U_{qq} = \frac{H(x_U + sq) + H(x_U - sq) - 2H(x_U)}{s^2}. \quad (5)$$

Thus these evaluations extract the action's exact cubic and quartic vertices, up to floating-point roundoff. They are not an extrapolated mass fit. Steps $s = 0.02$ and 0.01 supply an independent polynomial identity test. Hessians are evaluated by differentiating the actual potential and cached by its source, parameters and field coordinate.

2.2 Loop kernels and the hard-region scope

Use a complete orthonormal scalar basis with 55 heavy mass squares $m_i^2 > 0$. For the other 273 states set the propagator mass to zero *only in the leading retained-mass hard expansion*. The latter include the upper Landau Goldstones and PQ direction. They have not been removed from the state sum or declared physically massless by this approximation. For $x, y \geq 0$, at least one positive, define

$$A(x) = x \left(\log \frac{x}{\mu_0^2} - 1 \right), \quad (6)$$

$$b(x, y) = 1 - \frac{x \log(x/\mu_0^2) - y \log(y/\mu_0^2)}{x - y}, \quad (7)$$

$$S(x, y) = \int_0^1 \frac{t(1-t) dt}{(1-t)x + ty}. \quad (8)$$

The continuous limits are

$$b(x, x) = -\log(x/\mu_0^2), \quad b(x, 0) = 1 - \log(x/\mu_0^2), \quad S(x, x) = \frac{1}{6x}, \quad S(x, 0) = \frac{1}{2x}. \quad (9)$$

Expand $\frac{1}{2} \text{tr} \log[-\partial^2 + H(X)]$ to second order in external scalars. The seagull comes from the term linear in U and the bubble from the term quadratic in T . With $\sum'$ denoting pairs with at least one upper heavy line, the renormalized hard coefficients are

$$t_a^S = \frac{1}{2\mathcal{L}} \sum_{i \in h} A(m_i^2) T_{a,ii}, \quad (10)$$

$$\Pi_{ab}^S = \frac{1}{2\mathcal{L}} \left[\sum_{i \in h} A(m_i^2) U_{ab,ii} - \sum'_{ij} T_{a,ij} T_{b,ji} b(m_i^2, m_j^2) \right], \quad (11)$$

$$k_{ab}^S = \frac{1}{2\mathcal{L}} \sum'_{ij} T_{a,ij} T_{b,ji} S(m_i^2, m_j^2). \quad (12)$$

These signs follow directly from the second variation of the logarithm; the momentum derivative of the negative Euclidean bubble is positive. Pure-soft contributions are scaleless in this hard

expansion and belong to the common EFT subtraction. No retained-mass error bound follows from dropping them at this order.

For any real q , $q^T k^S q$ is a sum of nonnegative squares with weights $S > 0$. Hence $k^S \succeq 0$ is an independent analytical check, not a numerical assumption. The mass insertion has no such positivity theorem.

2.3 Reconstructing all 248 real directions by symmetry

The retained real representations are $H_8, F_{120}, L_{60}, R_{60}$. Each of H and F is a complex field in a real-type PS representation, and therefore consists of two equivalent real copies. A real symmetric PS-invariant bilinear has three coefficients on each such isotypic component. The complex-type irreducible L and R each contribute one. Thus the invariant space has dimension $3 + 3 + 1 + 1 = 8$.

Construct an orthonormal basis B_A of this commutant using the existing actual conjugate-field intertwiners, not arbitrary matrices fitted to eigenvalues. Choose eight directions for which $P_{iA} = q_i^T B_A q_i$ is invertible. For a calculated quadratic form F ,

$$f_A = (P^{-1})_{Ai} F(q_i, q_i), \quad F = \sum_A f_A B_A. \quad (13)$$

Two independent mixed-parent/complex-phase directions and all PS generator commutators verify the reconstruction. The scalar kinetic eigenvalues span

$$9.38883 \times 10^{-5} \leq \lambda(k^S) \leq 0.00450338, \quad (14)$$

whereas Π^S spans $[-0.1408203, -0.0117783]\omega^2$. Scalar-leg canonicalization contributes $\delta Y^a = -\frac{1}{2}k_{ab}^S Y^b$, including the allowed conjugate H/F invariants. It is a determined correction to the two UV spurions, not two extra free family matrices. It must not also be included as an external heavy leg when the same kernel is subsequently inverted.

3 One tadpole prescription across both backgrounds

3.1 Why the upper tadpoles cannot be reset independently

The previous broken-vacuum calculation fixed the bosonic finite mass counterterms to

$$(\delta\mu^2, \delta\nu^2, \delta\mu_s^2)/\omega^2 = (-0.05633071, -5.46974409, -0.03423771). \quad (15)$$

In canonical real coordinates their Hessian is

$$\mathbf{H}_{\text{CT}} = \text{diag} \left(-\delta\mu^2 I_{54}, -\frac{\delta\nu^2}{2} I_{252}, 0_{20}, -\delta\mu_s^2 I_2 \right). \quad (16)$$

The zero 10_H entry does not assert that the final single-light-state retuning is zero; that further operation has not been included here. At $\sigma_U = 0$ its tadpole cannot independently determine a second $\delta\nu^2$. Setting both backgrounds' tadpoles to zero by independently changing the same counterterms would silently change the prescription.

Let R contain the two radial field directions $\partial X/\partial w$ and $\partial X/\partial v_s$, which span the PS-invariant heavy tadpoles. The hard vector contribution follows by differentiating $3 \text{tr}[M_V^4(\log(M_V^2/\mu_0^2) - 5/6)]/(64\pi^2)$:

$$t_r^V = \frac{3M_V^2}{2\mathcal{L}} \left(\log \frac{M_V^2}{\mu_0^2} - \frac{1}{3} \right) \text{tr}(M_V^2)_{,r}. \quad (17)$$

For the common hard-plus-counterterm radial insertion,

$$t_r = t_r^S + t_r^V + (R^T \mathbf{H}_{\text{CT}} x_U)_r, \quad \delta r = -(R^T \mathbf{H} R)^{-1} t. \quad (18)$$

The actual values are

Contribution	w direction	v_s direction
Hard scalar tadpole	-0.0220062	-0.00210402
Hard vector tadpole	-0.0151893	0
Common counterterm	0.1353033	0.00870589
Residual t	0.0981077	0.00660187
Linear response δr	-0.0139319	-0.0555909

Tadpoles and responses carry ω^3 and ω , respectively. The v_s response is about 22% of $v_{s,U}$ and is not automatically a negligible background correction. Equation (18) contains only the hard-plus-CT part: retained EFT-loop tadpoles and fermionic fixed-VEV counterterms are still absent. It is *not* a completed upper quantum stationary solution.

3.2 Tadpole insertion from the tree heavy valley

Let y denote retained coordinates, z massive coordinates and write the tree solution as $z = z_{\text{cl}}(y)$. At the PS point the mixed tree Hessian between these inequivalent irreducible components vanishes, so $z_{,a} = 0$. Differentiating $V_{,z}(y, z_{\text{cl}}) = 0$ twice yields

$$z_{,ab} = -C^{-1}V_{,zab}, \quad C = V_{,zz} > 0. \quad (19)$$

Apply the chain rule to $V_{1,h}(y, z_{\text{cl}}) + V_{\text{CT}}(y, z_{\text{cl}})$:

$$\Delta D_{ab} = \Pi_{ab}^S + \Pi_{ab}^V + (H_{\text{CT}})_{ab} + t_z^T z_{,ab}, \quad (20)$$

$$t_z^T z_{,ab} = \delta z^T V_{,zab}, \quad \delta z = -C^{-1}t_z. \quad (21)$$

This derives the factorizable tadpole mass term without postulating an independently retuned upper vacuum. The two radial directions suffice for t_z by unbroken PS invariance. For the heavy six-mode propagator below, apply the same singlet-valley differentiation while holding those six modes as external coordinates; do not eliminate an external field twice. For active fields the linear vector-mass derivative vanishes, leaving

$$\Pi^V = \frac{3g_U^2 M_V^2}{\mathcal{L}} \left(\log \frac{M_V^2}{\mu_0^2} - \frac{1}{3} \right) C_S. \quad (22)$$

Combining the scalar and vector kinetic terms, the canonical mass increment is

$$\Delta D_{\text{can}} = \Delta D - \frac{1}{2} \{k^S + k^V, D_0\}. \quad (23)$$

It is real symmetric; positivity of the kinetic metric is checked. The current canonical mass increment ranges approximately from -0.0260 to $2.596\omega^2$. A small kinetic correction alone therefore does not justify perturbative control of the mass or the vacuum.

4 Factorizable upper scalar-exchange Wilson response

4.1 All Yukawa-coupled heavy scalar copies

Of the 55 positive upper scalars, the actual two UV Yukawa tensors couple to 24 real modes: four real six-dimensional PS copies. The symmetric invariant mass block has dimension $\dim \text{Sym}^2 \mathbb{R}^4 = 10$. The implementation solves

$$R_i^A X_{ij} - X_{ij} R_j^A = 0 \quad (24)$$

for the intertwiners between the six-dimensional copies using all PS generators. Ten quadratic probes reconstruct their full hard scalar mass block, including off-diagonal copy mixings; a separate

direction checks it. Internal scalar sums still run over all 328 states as in Eq. (12). Vector curvature, the common bosonic counterterms and the hard valley response are then added.

Choose real Hermitian current slots I from the actual fermion-pair vertices and use

$$S_2 = \frac{1}{2}\chi^T D\chi - \chi^T JI - \frac{1}{2}I^T CI. \quad (25)$$

Our 18 slots are the real and imaginary parts of nine pairs, including different-colour diquarks. They are probes of the scalar response, not a complete Lorentz/gauge four-fermion operator basis. Gaussian elimination gives the scalar-exchange contact kernel

$$C_0^{\text{exch}} = J_0^T D_0^{-1} J_0, \quad (26)$$

$$\Delta C^{\text{exch}} = \Delta J^T D_0^{-1} J_0 + J_0^T D_0^{-1} \Delta J - J_0^T D_0^{-1} \Delta D D_0^{-1} J_0. \quad (27)$$

Here ΔJ includes the previously computed pure-Yukawa and upper vector proper vertices and fermion external legs. Heavy scalar canonicalization is not also inserted into J : its compensation is already in D . Explicitly, under an infinitesimal real change of heavy field coordinates,

$$\Delta D \mapsto \Delta D + s^T D_0 + D_0 s, \quad \Delta J \mapsto \Delta J + s^T J_0, \quad (28)$$

the two added vertex terms in Eq. (27) cancel the two added propagator terms exactly. Finite differences of the full resolvent verify Eq. (27) independently.

4.2 A non-small insertion hidden by projected channels

For synthetic one-family choices $(h_{\text{raw}}, f_{\text{raw}}) = (0.20 + 0.04i, 0.025 - 0.01i)$ and $(0.13 - 0.07i, 0.041 + 0.013i)$, the ratios $\|\Delta C^{\text{exch}}\|_F / \|C_0^{\text{exch}}\|_F$ are approximately 0.0441 and 0.361. Neither is a fitted amplitude or proton lifetime. More importantly, define

$$E = D_0^{-1/2} \Delta D D_0^{-1/2}. \quad (29)$$

The actual computed subset has

$$\lambda(E) \in [-0.1314683, 6.7559262], \quad \rho(E) = 6.7559262. \quad (30)$$

On its dominant unit eigenvector u , the contractions $u^T D_0^{-1/2} \Delta D_X D_0^{-1/2} u$ from scalar loops, vector loops, the common bosonic counterterm and the valley response are respectively $(-0.538946, -0.016987, +7.346374, -0.034515)$. These add to 6.755926; they are not eigenvalues of the separate contributions. The large direction is therefore dominated by the transported finite counterterm, not a loss of numerical precision in the loop kernels.

Proposition 1 (Exact convergence criterion for this subset). *For $D_0 > 0$ and real symmetric ΔD , the Taylor/Neumann series of $(D_0 + \ell \Delta D)^{-1}$ about $\ell = 0$ converges in matrix norm if $|\ell| \rho(E) < 1$, and fails when $|\ell| \rho(E) > 1$.*

Proof. Factor the inverse as $D_0^{-1/2} (I + \ell E)^{-1} D_0^{-1/2}$ and orthogonally diagonalize E . Each eigenvalue gives an ordinary geometric series with ratio $-\ell \lambda(E)$. The largest absolute eigenvalue sets the radius of convergence. $\square$

The radius is about 0.1480, so the displayed subset is not a small insertion at $\ell = 1$. Positivity of $I + E$ does not repair convergence; both properties can and here do coexist. Small source projections can miss the large-eigenvalue directions and cannot certify the full kernel.

This is a precise obstruction to promoting the present fixed-parameter subset, not a proof of nonperturbativity or exclusion of the complete theory. Missing finite diagrams, fermionic counterterms, parameter feedback and retained-mass corrections may modify E . Resumming only this subset is not a new physical prediction. Nonfactorizable boxes, direct vector exchange, scalar three/four-point Wilson vertices and a complete operator basis remain uncomputed.

5 Lower constant-background finite scalar subtraction

At the actual broken background, let B_A span the 248 upper retained coordinates and B_H the 55 upper heavy coordinates. Write

$$A = B_A^T H B_A, \quad C = B_H^T H B_H, \quad W = B_H^T H B_A, \quad S_0 = A - W^T C^{-1} W. \quad (31)$$

With O the 24 upper gauge-orbit directions, the horizontal projector is $P = I - O(O^T O)^{-1} O^T$. At this background $B_H^T O = 0$, so in the 303-dimensional chart (B_A, B_H) the scalar kinetic metric is $\text{diag}(G_0, I_{55})$ with $G_0 = B_A^T P B_A$. The PQ spectator is not part of this chart. The exact scalar Schur kernel at Euclidean $z = p_E^2$ is

$$S(z) = A + zG_0 - W^T(C + zI)^{-1}W, \quad (32)$$

$$S_{[2]}(z) = S_0 + zG, \quad G = G_0 + W^T C^{-2} W. \quad (33)$$

Block Gaussian elimination gives the pointwise identity

$$\log \det(M + zG) = \log \det(C + zI) + \log \det S(z). \quad (34)$$

The full generalized scalar spectrum has 290 positive modes and 13 zeros; the local 248-dimensional operator has 235 positive modes and 13 zeros. The zeros include nine lower gauge directions and four light-Higgs real directions. They are not thirteen physical massless particles. The 290 positive masses agree with the original real Hessian.

Let

$$v(x; \mu_0) = \frac{x^2}{64\pi^2} \left(\log \frac{x}{\mu_0^2} - \frac{3}{2} \right), \quad v(0) = 0. \quad (35)$$

Denote the generalized full, upper- C , and local spectra by m_f^2 , m_C^2 and m_l^2 . In dimensional regularization, field-independent kinetic normalization factors multiply scaleless integrals and drop out. Integrating Eq. (34) in this scalar sector yields

$$V_{\text{PS,nonlocal}}^{(1)} = \sum_f v(m_f^2) - \sum_C v(m_C^2), \quad (36)$$

$$\Delta V_{\text{loc}}^{(1)} = \sum_f v(m_f^2) - \sum_C v(m_C^2) - \sum_l v(m_l^2). \quad (37)$$

Numerically, in ω^4 units,

$$V_{\text{full}} = -0.04307001295, \quad V_C = -0.00596443931, \quad (38)$$

$$V_{\text{PS,nonlocal}} = -0.03710557365, \quad V_{\text{PS,local}} = -0.03705904935, \quad (39)$$

$$\Delta V_{\text{loc}}^{(1)} = -4.65242986 \times 10^{-5}. \quad (40)$$

Six momentum checks verify Eq. (34); a transformed broken PS background gives the same finite difference. Its scale derivative is independently fixed by

$$\frac{d\Delta V_{\text{loc}}^{(1)}}{d \log \mu} = - \frac{\sum_f m_f^4 - \sum_C m_C^4 - \sum_l m_l^4}{32\pi^2}. \quad (41)$$

The conclusion is not that this vacuum constant is an observable or a complete lower threshold. It is that replacing the exact nonlocal operator by its two-derivative expansion *inside an unrestricted loop integral* loses a definite finite term. Higher-derivative/tree Wilson insertions and the associated subtraction must be tracked. Background/source derivatives, transverse gauge fields, ghosts, Goldstone gauge-fixing terms and finite Yukawa/box contributions remain necessary for the full gauge-covariant matching functional. The quotient calculation alone is not that functional.

6 Finite seesaw: complete restricted limit and actual subset

6.1 Conventions and the terminal canonical limit

We use $Y_{\alpha i}$ with lepton rows, $M_R^T = M_R$, and

$$U^T M_R U = \text{diag}(M_i), \quad M_i > 0, \quad Q = Y M^{-1} Y^T, \quad C_{5,\text{tree}}^I = -Q. \quad (42)$$

Our C_5 sign is opposite to that of Ref. [3]; our g_1 is GUT normalized, $g_Y^2 = 3g_1^2/5$. Write $\ell_i = \log(\mu^2/M_i^2)$. The canonical type-I one-loop result can be organized into the following finite contributions [3]:

$$z_H = \frac{1}{\mathcal{L}} \sum_i \|Y_i\|^2 (\tfrac{1}{2} + \ell_i), \quad (43)$$

$$k_{L,\text{row}} = \frac{1}{\mathcal{L}} Y \text{diag} \left(\frac{3 + 2\ell_i}{4} \right) Y^\dagger, \quad (44)$$

$$\delta C_g = -\frac{1}{\mathcal{L}} \sum_i \left[2\lambda(1 + \ell_i) + \frac{g_Y^2 + g_2^2}{4} (1 + 3\ell_i) \right] \frac{Y_i Y_i^T}{M_i}, \quad (45)$$

$$\delta C_{\text{legs}} = -z_H C_{5,\text{tree}}^I - \frac{1}{2} (k_{L,\text{row}} C_{5,\text{tree}}^I + C_{5,\text{tree}}^I k_{L,\text{row}}^T). \quad (46)$$

The row tensor is the transpose of the literal left-Weyl kinetic matrix; this distinction matters under complex family transformations. The terminal canonical result is $C_5 = -Q + \delta C_g + \delta C_{\text{legs}}$. Its applicability here is restricted to all sterile fields removed, $C_{HN} = 0$, no pre-existing C_5 , and leading $q_H/M_i^2 = 0$. It is a complete C_5 formula in that limit, not complete SMEFT matching. At $\mu = M$ the independent degenerate wave-function calculation gives the factors 1/2 and 3/4 [4].

6.2 Independent RG and covariance checks

At fixed parameters let $A_Y = Y Y^\dagger$. Differentiating the finite terms gives

$$\frac{d\delta C_g}{d \log \mu} = -\frac{4\lambda + \frac{3}{2}(g_Y^2 + g_2^2)}{\mathcal{L}} Q, \quad (47)$$

$$\frac{d\delta C_{\text{legs}}}{d \log \mu} = \frac{2 \text{tr}(A_Y) Q + \frac{1}{2}(A_Y Q + Q A_Y^T)}{\mathcal{L}}. \quad (48)$$

These expressions equal the independently implemented $\beta_{C_5}^{\text{below}} - \beta_{(C_5-Q)}^{\text{above}}$ in the canonical limit. In the active theory,

$$\beta_{C_5-Q} = \beta_{C_5} - \beta_Y M^{-1} Y^T - Y M^{-1} \beta_Y^T + Y M^{-1} \beta_M M^{-1} Y^T. \quad (49)$$

Keeping the matrix ordering is essential. Finite differences in $\log \mu$ test the relation without differentiating eigenvectors. Under $Y \mapsto U_L^T Y U_N$, $M \mapsto U_N^T M U_N$, the result transforms as $C_5 \mapsto U_L^T C_5 U_L$. Exactly degenerate positive Takagi blocks permit a real orthogonal residual transformation, under which every block sum is invariant. These tests verify generic complex families and an independent $O(3)$ block rotation.

6.3 What can be applied to the actual P54 trajectory

The actual P54 scalar elimination generates $C_{HN} \neq 0$ and a type-II Weinberg term. We explicitly reject applying the restricted terminal formula to that input. Instead, the new sequential callback adds only the universal δC_g of Eq. (45) for the removed Takagi block. The existing nonzero C_{HN} remains in running and the tree block-Schur pullback. No missing finite term is assigned zero and then labelled a complete result.

For $M_R = \begin{pmatrix} M_{rr} & M_{rh} \\ M_{hr} & M_{hh} \end{pmatrix}$, the exact tree map is

$$M'_r = M_{rr} - M_{rh} M_{hh}^{-1} M_{hr}, \quad (50)$$

$$Y'_r = Y_r - Y_h M_{hh}^{-1} M_{hr}, \quad C'_5 = C_5 - Y_h M_{hh}^{-1} Y_h^T. \quad (51)$$

Its field-dependent derivative gives the retained C_{HN} pullback. For a moving Takagi block, $M_{rh} = 0$ at the matching point but this covariant general formula is still the implementation's regression reference. For any removed block h , the finite gauge/quartic derivative is Eq. (47) with Q replaced by Q_h . It equals the gauge/quartic part of the corresponding active-theory beta difference.

The threshold event is solved from the running matrix,

$$\mu = \kappa s_{\max}(M_R(\mu)), \quad \kappa = 1 \quad \text{for the central diagnostic.} \quad (52)$$

Degenerate blocks are removed together. Event ordering is recalculated after each evolution and finite matching step, not frozen from the starting spectrum. Tests also vary κ to 0.8 and 1.25. The two existing synthetic families run from 0.1ω to $10^{-4}\omega$ through three live finite events. Relative terminal C_5 changes versus the old tree-threshold trajectories are 0.00221120 and 0.00224041. These are diagnostic changes, not fitted neutrino observables or calibrated error estimates. A finite Wilson coefficient need not be continuous: matching equates amplitudes, not raw coefficients.

Missing finite insertions of C_{HN} and pre-existing C_5 , mixed removed/retained sterile graphs, and correlated finite Y_ν, M_R , Higgs and wave-function maps prevent complete P54 finite seesaw closure.

7 Verification ledger and next decisions

Verifier (new unless indicated)	Checks passed
Upper scalar hard matching / common tadpole	14/14
Upper factorizable scalar-exchange Wilson	16/16
Lower finite scalar determinant subtraction	22/22
Finite seesaw limit and live universal subset	24/24
Original sequential-seesaw default regression	46/46

Passing these checks certifies the stated algebra and numerical tests, *not* diagram completeness, perturbativity or empirical validity. Each new JSON report records source hashes, numerical matrices and explicit false full-matching/fit flags. New polynomial vertices were computed only where the original cache lacked them; no new stationary point or lattice scan was performed.

The next work should address the structural obstruction before a fit:

1. Complete the common-prescription tadpole/parameter bookkeeping and remaining upper Wilson diagrams, including fermionic counterterms when actual family matrices are supplied. Separate source-projected smallness from the full mass-operator test in Eq. (30). If this benchmark remains uncontrolled, compare a consistently converted renormalization prescription or reject the benchmark; do not simply resum one subset or adjust a mass formula.
2. Differentiate the lower source-dependent hard-minus-EFT functional and include its gauge/ghost/Goldstone, Yukawa and contact diagrams. Use the exact scalar subtraction as a regression, not as a replacement for the missing functional. Bound the retained-mass expansion or keep the necessary mass dependence.
3. Complete finite seesaw in the actual EFT with C_{HN} , type-II C_5 and partially retained sterile fields, using the canonical limit and beta-difference identities as independent tests.

4. Only after matching and perturbative/background control, solve the same-prescription full complex light state, enforce one light-doublet condition and the heavy-block gap, then attempt the constrained fit.

The action remains frozen throughout this checkpoint. No physical-fit, determinant, portal or unconditional first-principles closure is promoted.

Reproducible artifacts

All paths below are relative to `route_f/`. Each verifier has an identically named JSON/Markdown pair in `output/`, without its `verify_` prefix:

- `code/verify_p54_upper_scalar_matching.py`;
- `code/verify_p54_upper_wilson_exchange.py`;
- `code/verify_p54_lower_scalar_finite_subtraction.py`;
- `code/verify_p54_finite_seesaw_matching.py`.

The first three accept `--cache-dir` pointing to the existing `tmp/p54_full_doublet_cw` cache in the original checkout. Newly needed polynomial Hessians are written to this worktree's `tmp/p54_upper_scalar_jets`. Runtime versions are NumPy 2.0.2, SciPy 1.13.1 and JAX/JAXlib 0.4.30. The original default seesaw regression was rerun; its tree-threshold report was refreshed with the current source hash while keeping its historical model/date label. It is distinct from the new finite-threshold report.

References

- [1] B. Henning, X. Lu and H. Murayama, One-loop matching and running with covariant derivative expansion, [arXiv:1604.01019](#).
- [2] M. Gabelmann, M. Mühlleitner and F. Staub, Automatised matching between two scalar sectors at the one-loop level, *Eur. Phys. J. C* **79**, 163 (2019).
- [3] D. Zhang and S. Zhou, Complete one-loop matching of the type-I seesaw model onto the Standard Model effective field theory, especially Eqs. 49, 51, 89 and 91, [arXiv:2107.12133v3](#).
- [4] T. Ohlsson and M. Pernow, One-loop matching conditions in neutrino effective theory, especially Eqs. 42 and 45, [arXiv:2201.00840v2](#).